

Stuck & Recovery Analysis

Endovascular navigation RL · v1b vs v1bp · explore-only

14,691

v1b episodes

4,055,067

v1b explore steps

8,040

v1bp episodes

2,551,498

v1bp explore steps

SCOPE. Every number here is measured from the worker-log complete ledger (EPISODE_START / STEP / EPISODE_OUTCOME) of two training runs, restricted to EXPLORE episodes. Evaluation episodes are excluded exactly: runner.eval() resets with explicit seeds, so eval EPISODE_START lines carry a `seed=` field and explore resets do not. That filter yields v1b 14,691 explore / 1,470 eval (= 15 evals x 98) and v1bp 8,040 / 980 (= 10 x 98) — exact on both runs. The obvious alternative, main.log time windows, leaks (it caught only 567 of v1bp's 980) and episode_summary.jsonl is unusable because it logs only ~36% of explore episodes. TRAINING-TIME AXIS. Cumulative explore steps, summed per-episode in chronological order. Exact from the logs and directly comparable across runs, unlike wall-clock (which conflates simulator speed) or the global step counter. THE RUNS. v1b and v1bp are byte-identical except for one change: v1bp adds a reward pair (tip-average progress + catheter-slack potential) ported from a second machine. This is a single-variable experiment.

Generated from monitoring/extract_stuck.py records · canonical detector config unless stated

Definitions — and why each one is drawn where it is

Every downstream number depends on these choices, so they are stated before any result

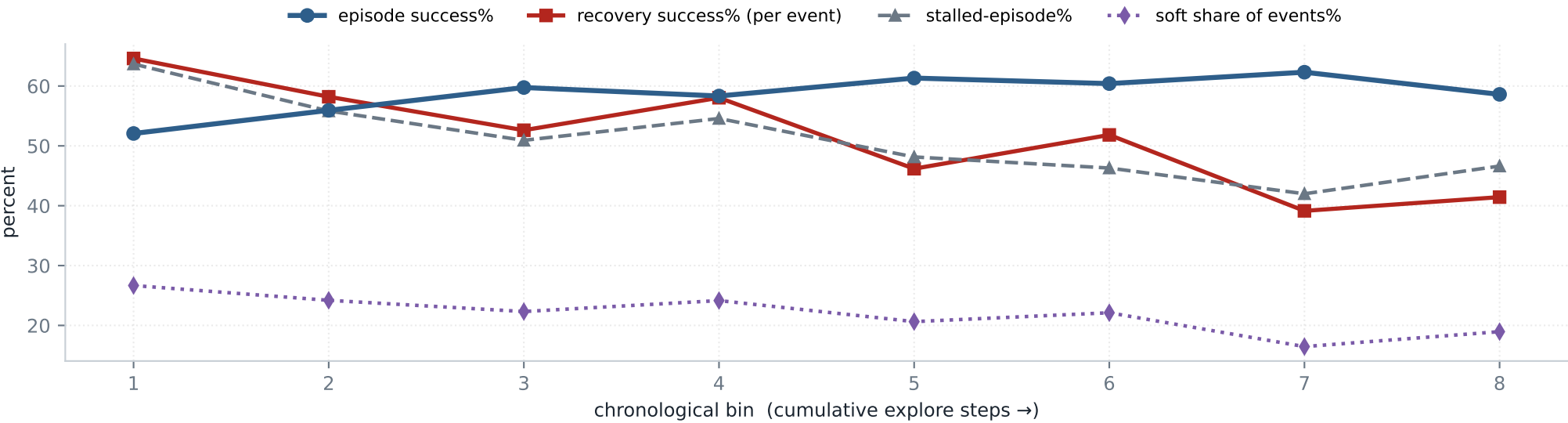
term	unit	definition
stall event	per event	wire pushing ($ gw_trans > 2$ mm/s) while path progress stays flat (< 0.3 mm gain) for 12 consecutive steps
grind	per event	escaped the stall after < 1 mm of withdrawal — forced through
soft	per event	escaped after 1–8 mm withdrawal — ease slack, re-advance (the manoeuvre we set out to induce)
hard	per event	escaped after > 8 mm withdrawal — costly pullback
unrecovered	per event	never passed the jam; terminal for that episode
recovery success%	per EVENT	(soft + hard + grind) / all stall events
episode success%	per EPISODE	reached the target

A stall is 'pushing but not progressing', not merely 'not progressing'. A wire that is deliberately retracting or rotating is manoeuvring, not stuck. Requiring the push command isolates the case where the policy IS trying to advance and the vessel will not allow it. The stall counter decays by 2 per non-stalled step rather than resetting, so one noisy command sign-flip inside a genuinely stuck push does not erase the evidence. Recovery kind is classified by ACTUAL withdrawn length — peak insertion minus deepest insertion during the stall — not by the commanded retraction. A retract command that moves nothing (wire wedged) must not score as a soft recovery. THE THREE SUCCESSES sit at different units of analysis, which is the single most important thing on this page: (1) EPISODE success is per episode — did it reach target. (2) RECOVERY success is per EVENT — did the wire escape a given stall. (3) RECOVERY-TO-EPISODE success is per episode — did escaping actually convert into reaching the target. Conflating (1) and (2) makes 'success tracks recovery' look true when the opposite holds.

T1 — v1b: stuck & recovery vs training time

8 equal-episode chronological bins · explore only · canonical detector

bin	explore steps (k)	eps	ep succ%	stalled-ep%	events	evt/stalled-ep	recovery succ%	soft%	hard%	grind%	unrec%
1	0-603	1836	52%	64%	2441	2.1	65%	27%	11%	27%	35%
2	604-1148	1836	56%	56%	1926	1.9	58%	24%	14%	20%	42%
3	1149-1647	1836	60%	51%	1568	1.7	53%	22%	12%	18%	47%
4	1647-2157	1836	58%	55%	1797	1.8	58%	24%	12%	21%	42%
5	2157-2629	1836	61%	48%	1323	1.5	46%	21%	8%	17%	54%
6	2629-3108	1836	60%	46%	1501	1.8	52%	22%	11%	18%	48%
7	3108-3559	1836	62%	42%	1142	1.5	39%	16%	10%	13%	61%
8	3559-4053	1836	59%	47%	1308	1.5	41%	19%	6%	17%	59%



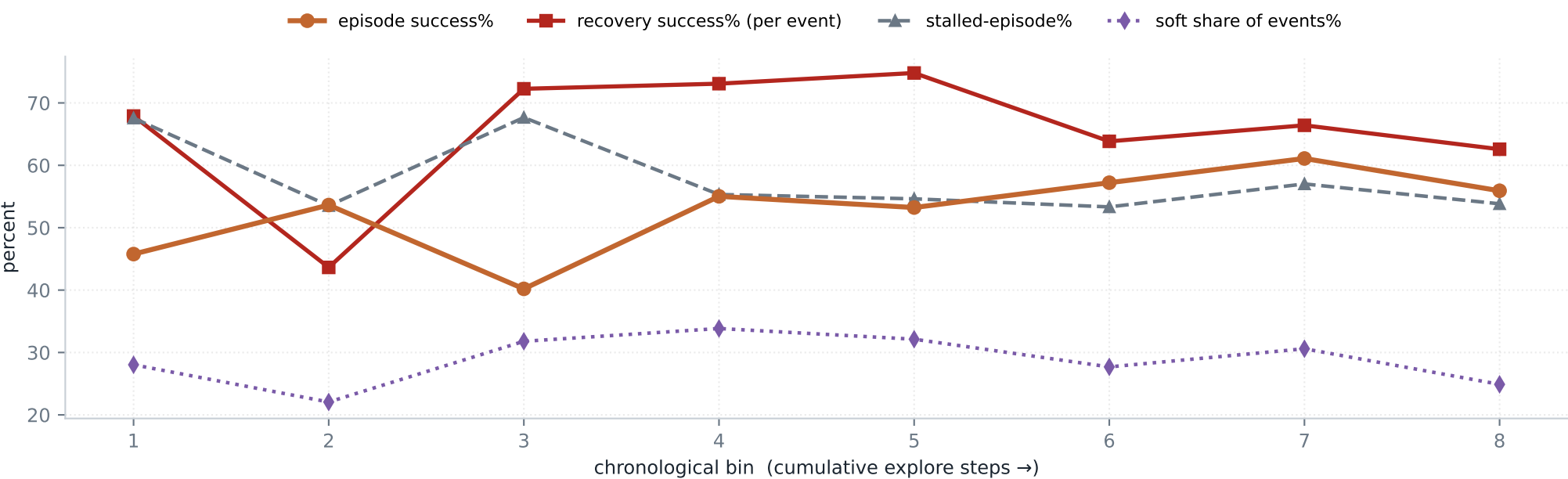
WHAT IT SHOWS How often the wire got stuck, how often it got out, and by which manoeuvre, as training progressed. Read the trend across bins, not the absolute levels.

INFERENCE The central surprise of the whole analysis. Episode success CLIMBS 52→62% while recovery success FALLS 65→39% and unrecovered rises 35→61%. What is actually improving is the stalled-episode share, which drops 64→42%. The policy is not learning to escape stalls — it is learning not to enter them, and the escape skill withers as stalls leave its training data. That is textbook interference: the skill starves because success removes its own training signal.

T1 — v1bp: stuck & recovery vs training time

8 equal-episode chronological bins · explore only · canonical detector

bin	explore steps (k)	eps	ep succ%	stalled-ep%	events	evt/stalled-ep	recovery succ%	soft%	hard%	grind%	unrec%
1	0-374	1005	46%	68%	1623	2.4	68%	28%	14%	26%	32%
2	375-671	1005	54%	54%	825	1.5	44%	22%	11%	11%	56%
3	671-1060	1005	40%	68%	1947	2.9	72%	32%	22%	18%	28%
4	1061-1370	1005	55%	55%	1501	2.7	73%	34%	18%	21%	27%
5	1370-1686	1005	53%	55%	1603	2.9	75%	32%	21%	22%	25%
6	1686-1979	1005	57%	53%	1181	2.2	64%	28%	17%	19%	36%
7	1979-2257	1005	61%	57%	1170	2.0	66%	31%	15%	21%	34%
8	2257-2551	1005	56%	54%	1149	2.1	63%	25%	15%	22%	37%



WHAT IT SHOWS The same trajectory for the run that added the reward pair (tip-average progress + catheter-slack potential).

INFERENCE The opposite internal dynamic. Recovery success does NOT decay — it holds 63–75% from bin 3 onward, against v1b's decay to ~40%. The reward pair kept the recovery skill alive, exactly as designed. But note the episode-success column is no better for it (46→56%), which is the tension resolved in T5.

T2 — the three successes: where episode success actually comes from

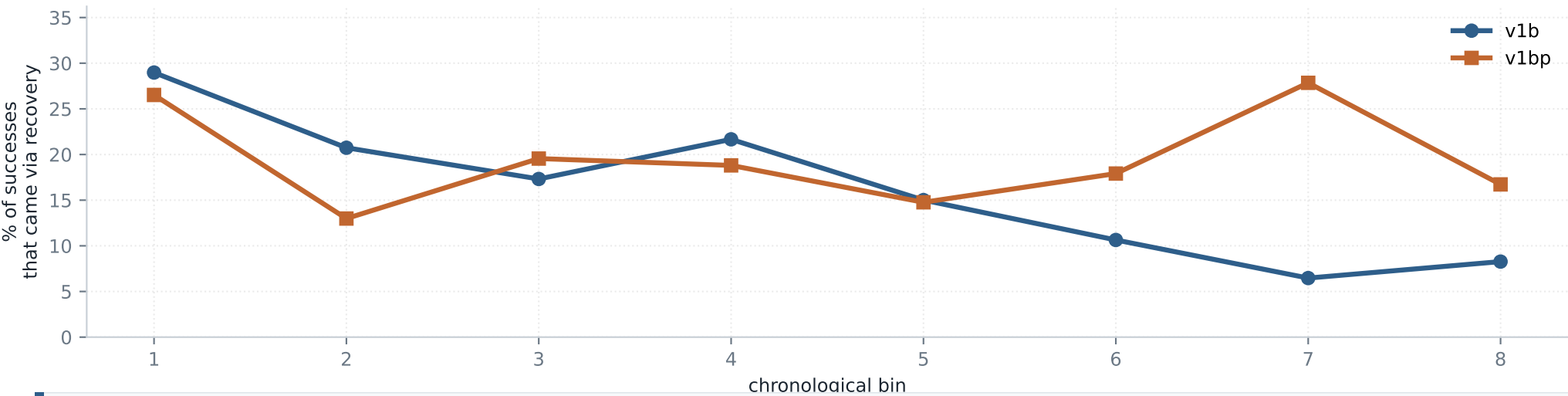
Each episode assigned to exactly one channel: never stalled / stalled-and-escaped / stalled-and-stuck

v1b

bin	eps	ep succ%	clean eps (succ%)	covered eps (succ%)	rec eps (succ%)	successes via rec
1	1836	52%	667 (100%)	305 (91%)	864 (1%)	29%
2	1836	56%	811 (100%)	220 (97%)	805 (0%)	21%
3	1836	60%	901 (100%)	192 (99%)	743 (1%)	17%
4	1836	58%	834 (100%)	248 (94%)	754 (1%)	22%
5	1836	61%	952 (100%)	172 (98%)	712 (1%)	15%
6	1836	60%	986 (100%)	127 (93%)	723 (1%)	11%
7	1836	62%	1065 (100%)	76 (97%)	695 (1%)	6%
8	1836	59%	980 (100%)	90 (99%)	766 (1%)	8%

v1bp

bin	eps	ep succ%	clean eps (succ%)	covered eps (succ%)	rec eps (succ%)	successes via rec
1	1005	46%	326 (100%)	158 (77%)	521 (2%)	27%
2	1005	54%	467 (100%)	73 (96%)	465 (0%)	13%
3	1005	40%	325 (100%)	140 (56%)	540 (0%)	20%
4	1005	55%	449 (100%)	152 (68%)	404 (0%)	19%
5	1005	53%	456 (100%)	145 (54%)	404 (0%)	15%
6	1005	57%	469 (100%)	109 (94%)	427 (1%)	18%
7	1005	61%	432 (100%)	180 (95%)	393 (3%)	28%
8	1005	56%	464 (100%)	111 (85%)	430 (1%)	17%



WHAT IT SHOWS Episode success decomposed by channel, and the share of all successes that flowed through recovery (the chart).

INFERENCE In v1b recovery's contribution COLLAPSES from 29% to 6%: early on a third of wins were rescues, by the end nearly every win is an episode that simply never jammed. v1bp shows no such decay (13–28%, and its best bin leans MOST on recovery). Two runs, nearly identical peak success, opposite routes: v1b avoids-and-runs-clean, v1bp stalls-and-fights-through. CAVEAT: the 100% and ~1% columns are near-definitional — an episode that pushes 600 steps without jamming has by construction arrived, one ending jammed has not. Read the COUNTS and the final column, not those rates.

T3 — does recovery TYPE predict episode success?

Episodes grouped by the best recovery they achieved · tests the premise behind the crunchpass / two-mode design

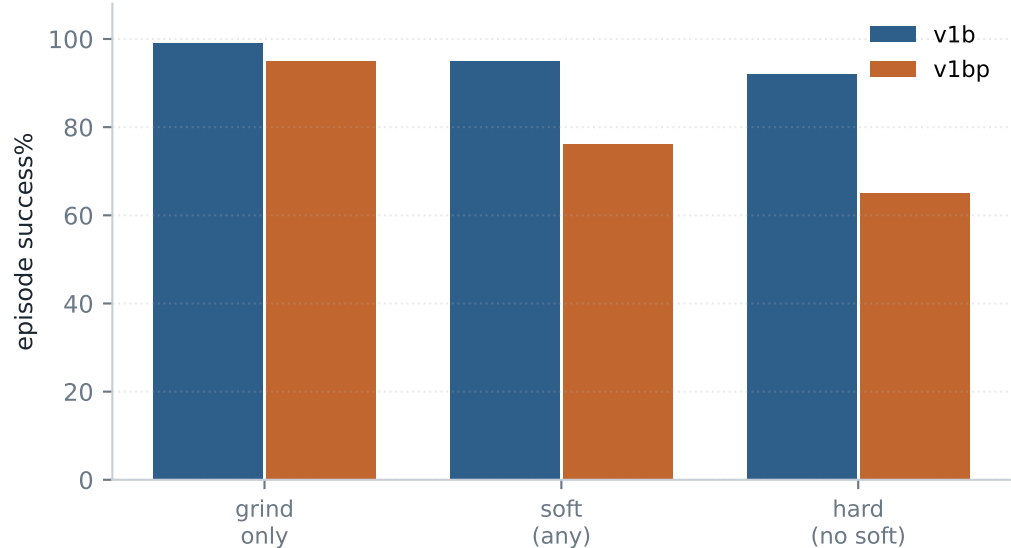
v1b

recovery profile	episodes	episode succ%	median step	mean retract m
soft (any)	820	95%	187	4.9
hard (no soft)	248	92%	227	13.7
grind only	362	99%	107	0.1
unrecovered (any)	6065	1%	600	13.5
never stalled	7196	100%	41	0.0

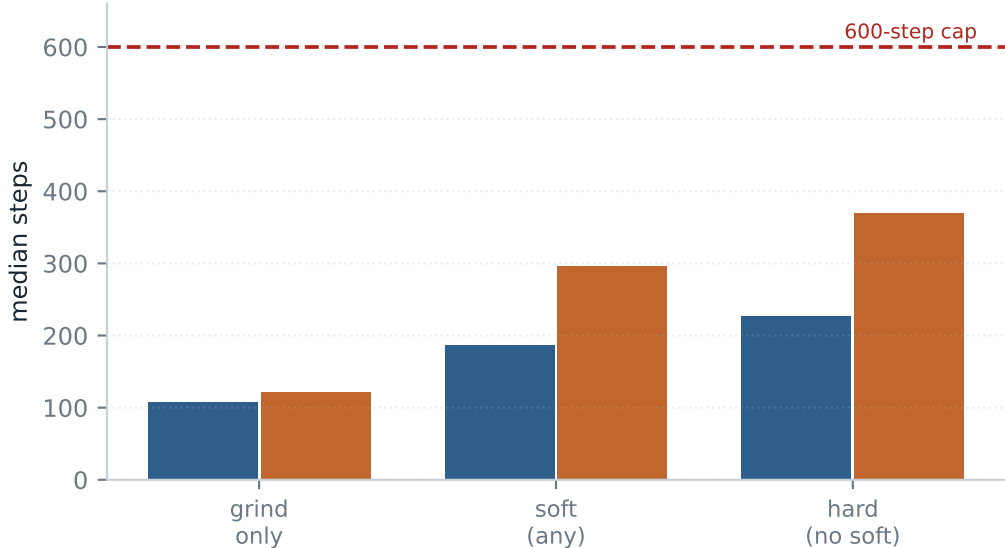
v1bp

recovery profile	episodes	episode succ%	median step	mean retract m
soft (any)	714	76%	296	6.2
hard (no soft)	195	65%	370	15.2
grind only	159	95%	121	0.1
unrecovered (any)	3584	1%	600	11.0
never stalled	3388	100%	43	0.0

episode success by recovery type



time cost of each recovery type



WHAT IT SHOWS Whether the type of escape predicts the episode outcome, and what each type costs in steps.

INFERENCE GRIND BEATS SOFT in both runs — 99% vs 95% (v1b) and 95% vs 76% (v1bp) — and does it in roughly a third of the steps. This inverts the premise behind the crunchpass lane and the 'we want soft recoveries' framing. The likely reason is a confound: recovery type is mostly a proxy for stall SEVERITY, not policy skill. A jam you can grind through was mild; one forcing an 8 mm withdrawal was severe and usually deeper, where the episode was already in trouble. Either way the practical conclusion holds — soft recovery cannot be justified as a training target on this evidence until the confound is broken by comparing within matched depth and severity strata.

T4 — stuck & recovery by depth

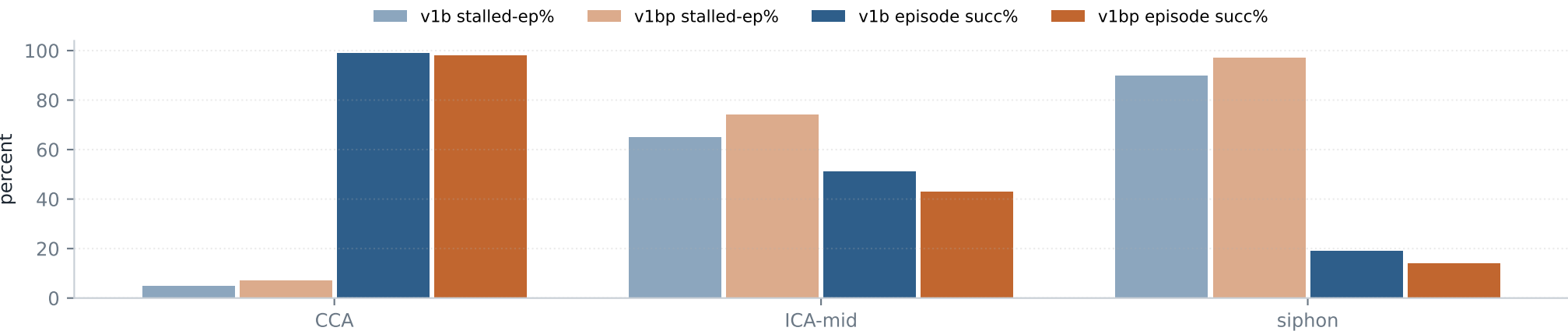
Path-length bands: CCA (<146 mm) · ICA-mid (146–210 mm) · siphon (>210 mm)

v1b

band	eps	ep succ%	stalled-ep%	events	recovery succ%	soft%	hard%	grind%	unrec%
CCA	5368	99%	5%	496	94%	43%	15%	36%	6%
ICA-mid	4692	51%	65%	5063	54%	23%	14%	18%	46%
siphon	4631	19%	90%	7454	50%	21%	9%	20%	50%

v1bp

band	eps	ep succ%	stalled-ep%	events	recovery succ%	soft%	hard%	grind%	unrec%
CCA	2804	98%	7%	457	86%	36%	18%	32%	14%
ICA-mid	2690	43%	74%	4217	65%	30%	17%	19%	35%
siphon	2546	14%	97%	6325	68%	29%	18%	21%	32%



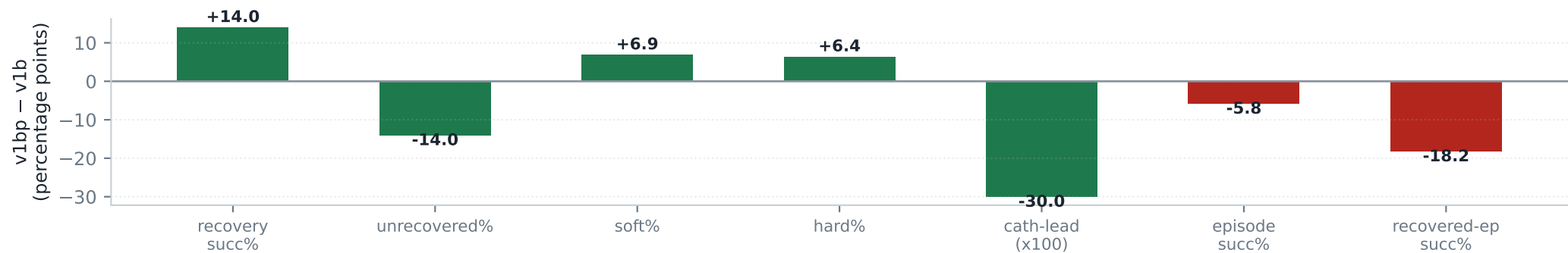
WHAT IT SHOWS Where the stalls actually are, and whether escaping them rescues the episode there.

INFERENCE Depth dominates every other factor. Stalling runs 5–7% at the CCA but 90% (v1b) to 97% (v1bp) at the siphon — getting stuck is essentially guaranteed once the wire enters the cavernous segment. And escaping does not save you there: v1bp escapes 68% of siphon stalls yet only 14% of siphon episodes succeed. You escape one jam and immediately meet the next. This reframes the siphon as a STALL-DENSITY problem rather than an escape-skill problem, and it is the same wall the real-patient evaluation hit (0 of 30), seen from the inside.

T5 — head-to-head: what the reward pair actually did

Whole run, canonical detector. v1bp = v1b + (tip-average progress + catheter-slack potential)

metric	v1b	v1bp	reading
episodes	14,691	8,040	
explore steps	4,055,067	2,551,498	
episode succ%	58.6	52.8	v1b ahead (57.4% at matched 2.55M budget)
stalled-ep%	51.0	57.9	v1bp jams MORE
stall events	13,013	10,999	
recovery succ%	53.4	67.4	v1bp escapes FAR better
soft%	22.6	29.5	
hard%	10.9	17.3	
grind%	19.8	20.6	
unrecovered%	46.6	32.6	v1bp leaves far fewer terminal jams
clean-ep succ%	100.0	100.0	
recovered-ep succ%	95.2	77.0	v1bp's escapes CONVERT WORSE
% successes via recovery	15.8	19.4	
mean cath-lead frac	0.34	0.04	the catheter shove is gone

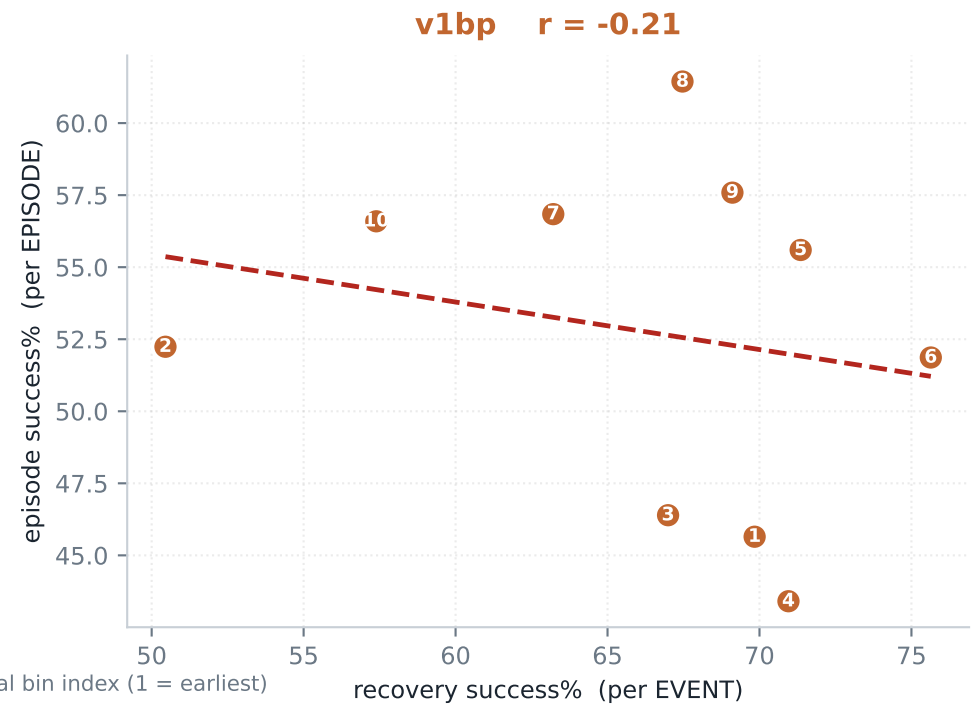
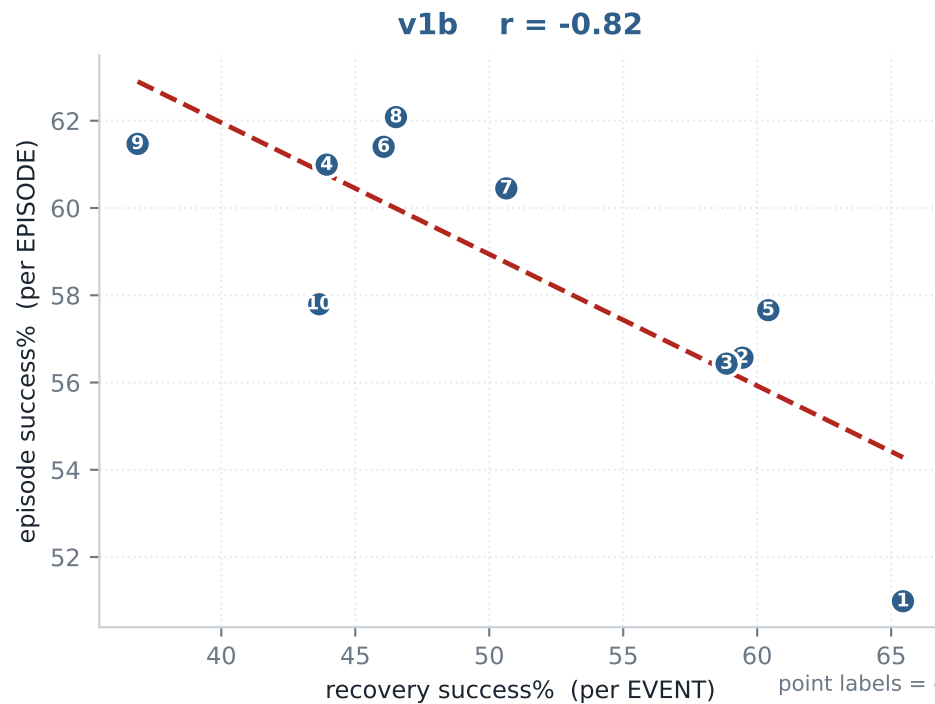


WHAT IT SHOWS The single-variable reward-pair experiment, aggregated over both whole runs.

INFERENCE A genuinely instructive SPLIT result. The pair did exactly what it was designed to do: catheter-lead — the shove pathology it targeted — fell 0.34→0.04, escape rose 53→67%, terminal jams fell 47→33%. And it gained nothing: episode success is LOWER (52.8 vs 58.6; 57.4 at matched budget). The mechanism is that v1bp's careful non-shoving gait jams more often (57.9 vs 51.0%) and its recoveries convert at only 77% vs 95%, because they are slow — soft recoveries take 296 median steps against a 600 cap. This is the cleanest evidence in the program that you can move an internal behaviour decisively and still gain nothing on the outcome, which is a result worth reporting in its own right.

T6 — is episode success actually tracking recovery?

10 chronological bins per run · each point is one bin



WHAT IT SHOWS A direct test of the intuition that success rises and falls with recovery skill. The dashed line is the least-squares fit across bins.

INFERENCE In v1b the relationship is **STRONGLY NEGATIVE** ($r = -0.82$): periods of better recovery are periods of worse success, because both are downstream of how often the wire is jamming at all. An earlier reading in this program that 'success% $\approx$ recovery%' was wrong — it compared a per-EVENT rate against a per-EPISODE rate without noticing they are different units, which is exactly why the three successes are separated in T2. v1bp's near-zero $r = -0.21$ is itself informative: the reward pair decoupled the two, which is what preserving the recovery skill looks like statistically.

T7 — threshold sensitivity: are any of these conclusions detector artifacts?

Every episode scored under three detector configurations simultaneously

v1b

config	events	stalled-ep%	recovery succ%	soft%	hard%	grind%	unrec%
sens	17,386	55%	65%	29%	10%	26%	35%
canon	13,013	51%	53%	23%	11%	20%	47%
strict	10,394	48%	42%	18%	11%	14%	58%

v1bp

config	events	stalled-ep%	recovery succ%	soft%	hard%	grind%	unrec%
sens	14,271	61%	75%	36%	16%	23%	25%
canon	10,999	58%	67%	30%	17%	21%	33%
strict	8,701	55%	60%	26%	18%	15%	40%

detector parameters

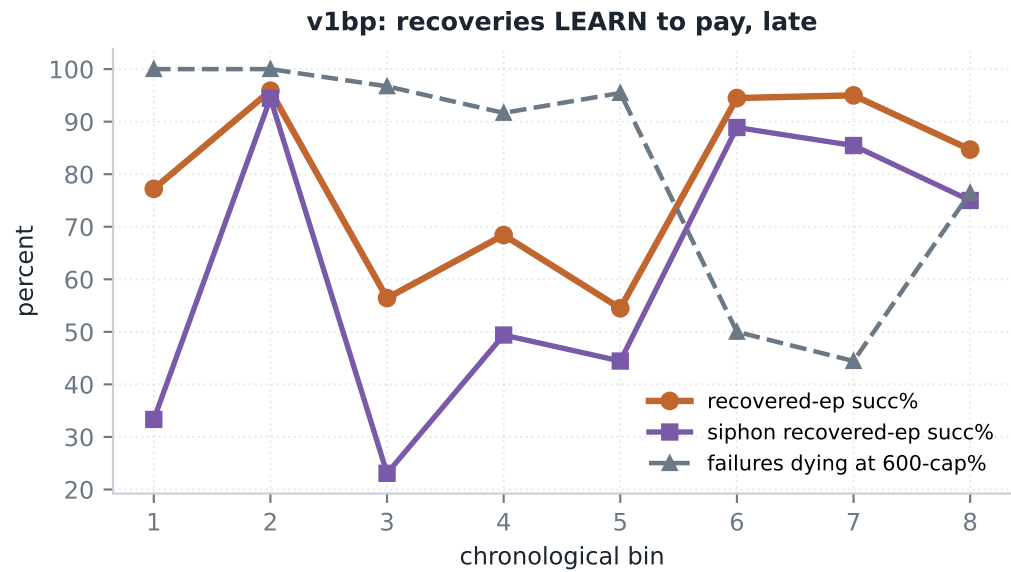
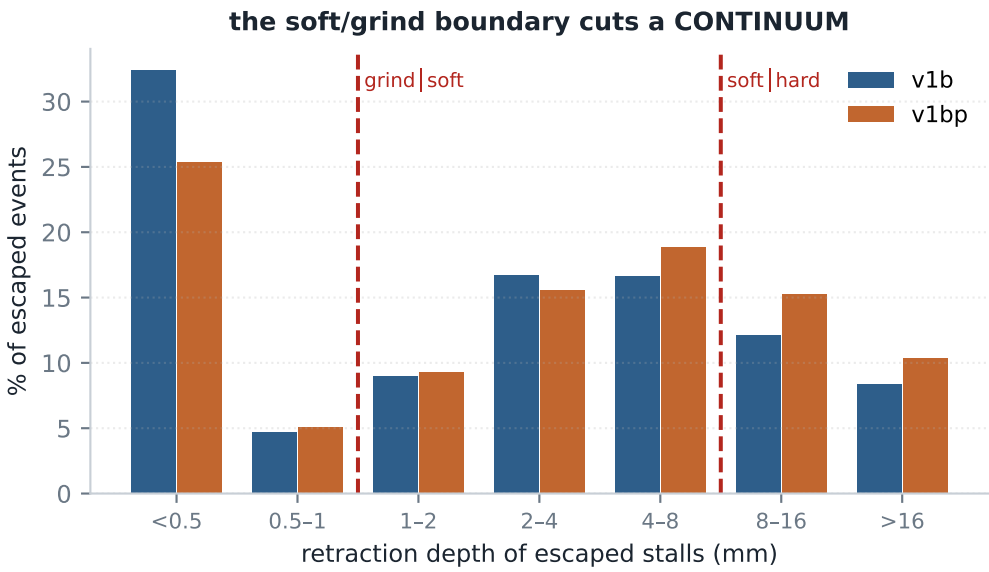
config	stall_eps	push_min	stuck_steps	retract_min	soft_max	pass_eps
sens	0.3 mm	1.0 mm/s	8	0.5 mm	8 mm	1.0 mm
canon	0.3 mm	2.0 mm/s	12	1.0 mm	8 mm	1.0 mm
strict	0.5 mm	4.0 mm/s	16	1.0 mm	8 mm	1.0 mm

WHAT IT SHOWS The same analysis re-run under a sensitive, a canonical, and a strict definition of 'stuck', to separate signal from detector choice.

INFERENCE Absolute rates move a great deal — v1b's recovery success reads anywhere from 42% to 65% depending on strictness — so NO absolute number here should be quoted on its own. But every ORDERING survives all three settings: v1bp beats v1b on escape rate and soft share and has fewer terminal jams, under sensitive, canonical and strict detectors alike. That is why every finding in this report is stated as a delta or a trend rather than a level.

Verification — reconciling with the earlier audit, and why v1bp's recoveries fail

Run on challenge: soft counts far exceeded a previous audit, and v1bp conversion fell as low as 54%



conversion gap is real within EVERY band

band	v1b rec-ep succ%	v1bp rec-ep succ%
CCA	98% (n=251)	94% (n=144)
ICA-mid	97% (n=733)	85% (n=503)
siphon	91% (n=446)	61% (n=421)

they time out, they do not jam

v1bp recovered-but-FAILED	value
died at 600-step cap	91% (211/246 max_steps)
median final distance to target	61 mm
last recovery at	97% of episode

WHAT IT SHOWS Left: why soft% is so sensitive to threshold choice. Right and below: the forensics of v1bp's recovered-but-failed episodes.

INFERENCE The retraction distribution is a CONTINUUM massed near zero (32% of v1b escapes are under 0.5 mm), not a bimodal soft/grind split — so the 1 mm and 8 mm cuts slice a smooth curve and absolute soft% is detector-defined. That fully explains the discrepancy with the earlier audit, whose stricter stall criterion admitted only deep, mostly hopeless stalls; no re-slicing of today's data reproduces its 2-6%, and the invariant orderings agree across both audits. Separately, v1bp's failed recoveries TIME OUT rather than jam (91% at the cap, 61 mm short, still fighting at 97% of the episode), the gap is real within every depth band, and it is largely TRANSIENT — by bins 6-7 conversion reaches 94-95%. Late v1bp did learn to make recoveries pay; what it never learned is avoidance.

Findings

Every claim below is stated as a delta or trend, and survives all three detector configurations

F1 Improvement is stall AVOIDANCE, not stall ESCAPE.

In v1b episode success rises 52→62% while recovery success falls 65→41%, unrecovered climbs 35→59%, and recovery's share of successes collapses 29→6%. The correlation is $r = -0.82$. The policy is not learning to get unstuck; it is learning not to get stuck, and the recovery skill decays as it does. This is the interference mechanism the design doc predicted, now measured.

F2 The reward pair fixed recovery — mechanically and unambiguously.

v1bp escapes 67.4% of stalls vs 53.4%, leaves 32.6% unrecovered vs 46.6%, raises soft share to 29.5%, and cuts catheter-lead from 0.34 to 0.04. The decay is gone: recovery holds 63-75% late ($r = -0.21$) where v1b decayed to ~40%.

F3 ...and it still did not help, because recovery does not pay.

Episode success is lower (52.8 vs 58.6; 57.4 at matched budget). Recovered episodes convert at 77% vs 95% and cost far more time (soft 296 vs 187 median steps against a 600 cap). v1bp also stalls more (57.9 vs 51.0%). More stalls, better escapes, worse conversion, slightly worse outcome — but by bins 6–7 conversion recovers to 94–95%, so the deficit was largely transient.

F4 GRIND beats SOFT on episode success, in both runs.

Grind-only episodes succeed 99%/95% in ~110 steps; soft 95%/76% in 187-296; hard 92%/65%. This inverts the premise behind the crunchpass lane. Likely a confound — recovery type proxies stall SEVERITY rather than policy skill — so soft recovery cannot be a training target until the confound is broken within matched depth and severity strata.

F5 Depth dominates everything.

90% (v1b) to 97% (v1bp) of siphon episodes stall, versus 5-7% at CCA. At the siphon, escaping does not save the episode: 50-68% escape but only 14-19% success. The limiting factor is stall DENSITY, not escape ability — the same wall the real-patient eval hit at 0 of 30.

F6 Detector-robust.

Absolute rates move ± 20 pp across the three configurations, but every v1b-vs-v1bp ordering holds under all of them.

Implications for what to do next

What the measurements license, and what they rule out

1 The crunchpass lane's premise needs revisiting BEFORE launch.

It resamples success-conditioned crunch passages on the theory that soft recovery is the skill to amplify. F4 says grind-through is what actually converts and F1 says the winning channel is avoidance. Amplifying recovery may be optimising a behaviour that does not pay. Break the severity confound first: compare soft vs grind within matched depth strata.

2 If recovery is still wanted, it must be made to PAY.

Recovered episodes cost 2–7× the steps of clean ones. Under a 600-step cap with a sparse target reward, a slow recovery is nearly worthless — the optimiser is CORRECT to prefer avoidance. Either the budget must accommodate recovery or recovery must get cheaper.

3 The reward pair is validated as a mechanism and rejected as a win.

Moving an internal behaviour decisively (cath-lead 0.34→0.04, unrecovered −14 pp) produced no outcome gain. This is strong evidence against further reward shaping as the lever, and it is a publishable negative result in its own right.

4 The siphon wall is not a recovery problem.

Near-total stall rates with ~65% escape but ~15% success mean the limiting factor is the sheer density of stalls, not the ability to escape any one of them. Interventions aimed at escape skill are aimed at the wrong target.

5 Two runs, two equilibria, one summit.

v1b avoids-and-runs-clean; v1bp stalls-and-fights-through. They reach nearly identical peak success (61-62%) by opposite routes. Whatever caps performance is above both strategies, which argues that the next gain will not come from tuning the gait.

Reproduce: `monitoring/extract_stuck.py <run_dir> <out.jsonl> then monitoring/analyze_stuck.py stuck_v1b.jsonl stuck_v1bp.jsonl [canon|sens|strict]`